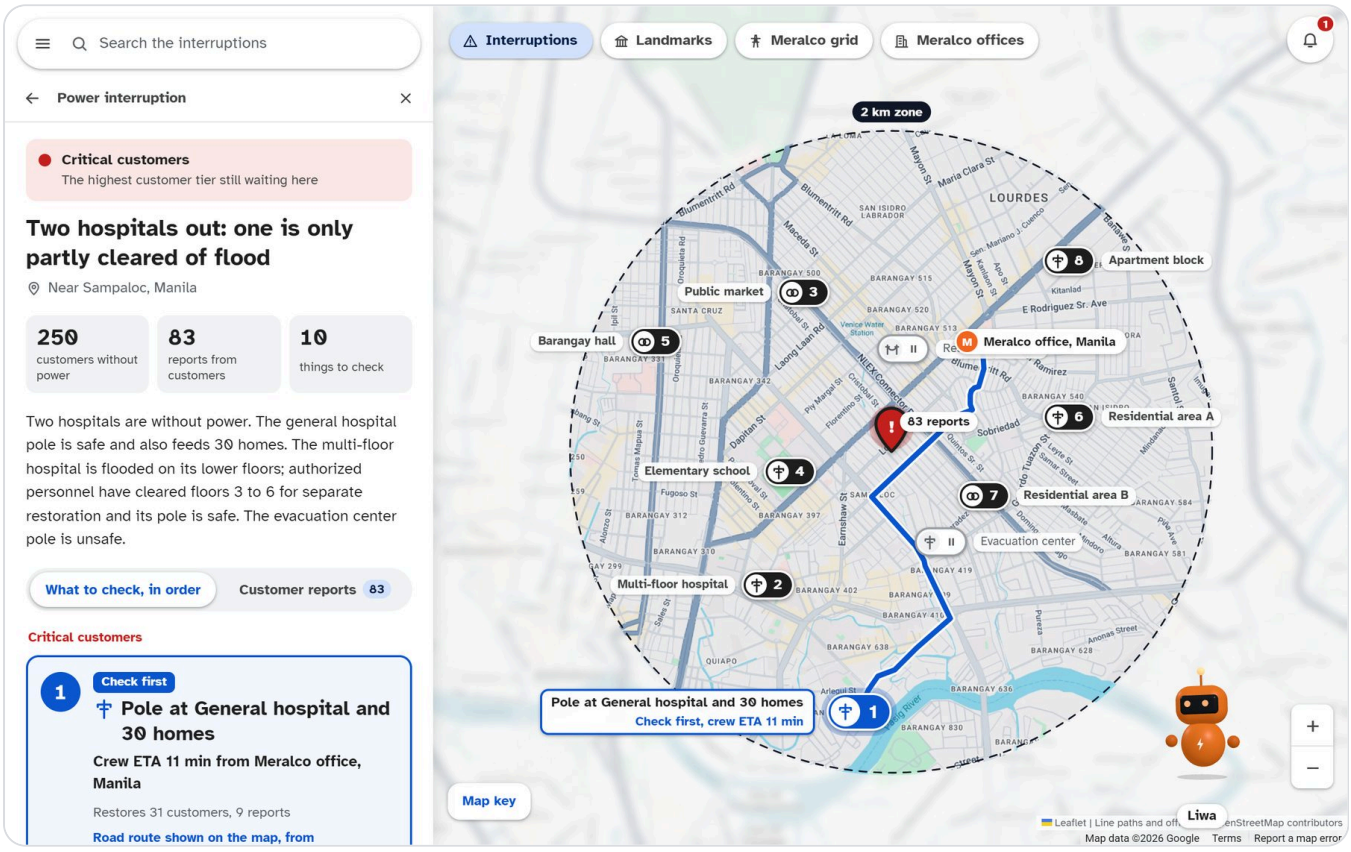
Team 403 Forbidden, Meralco IDOL Hackathon

FirstLook

How the system works

After a typhoon there are many power interruptions at once and too few crews. FirstLook shows each interruption on a map and tells the crew what to check first, what comes next, and why. This guide explains every screen, the rules behind the ranking, how Liwa (the planner) decides, and what to say when the panel asks.



Everything in FirstLook runs on simulated data: an invented power network over real places in Greater Manila, 20 test cases, invented customer reports and invented past typhoons. Nothing here is Meralco's own data.

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Words used in this guide. An **interruption** is one area without power. Inside it are pieces of **equipment** (poles, transformers, line sections), each one a **job** for the crew. **Number 1** is the job to check first. A job **on hold** cannot be worked on yet. **Liwa** is the planner that ranks the jobs. **Customers** means customer accounts, not people.

1 FirstLook on one page

Know where to look next. FirstLook recommends; people decide; Meralco's crews do the work.

The problem

When a typhoon passes, a distribution utility faces dozens of interruptions at the same time. Each one has several pieces of damaged equipment, and some of them feed hospitals or evacuation centers. Some sites are not safe to reach, and some are flooded. With one truck per area, someone has to decide where it goes first, then second, and do it the same way every time, under pressure.

What FirstLook does

- Puts every interruption on a map, grouped by the most important customers still waiting.
- Opens one interruption as a **2 km zone** and shows the equipment inside it, numbered in the order to check it: **1** first, then 2, 3 and so on. Jobs that cannot be worked on yet are shown as **on hold**, with the reason.
- Gives the crew's **ETA** from the nearest Meralco office and draws the **road route** to number 1.
- Explains every place in the order in plain words, and lists the **customer reports** behind it.
- Ranks again when **conditions change**, and keeps a **history** of past typhoons with **PDF reports**.

From typhoon to report

1
The typhoon passes
A simulated storm crosses Luzon. Nothing is shown until it has passed.

2
Interruptions appear
Each one is a pin on the map and a row in the side panel.

3
Open one
The map focuses on its zone. Liwa ranks its equipment in 6 to 10 seconds.

4
The crew goes to number 1
The route and ETA are shown. When the crew has checked it, press *Resolve*.

5
Number 2 is next
The route moves with it, and the count of customers without power drops.

6
Conditions change
A held site becomes safe or the flood is cleared. Liwa ranks again.

7
History
Past typhoons: what happened and how fast power came back.

8
PDF reports
A restoration report per typhoon, and a situation report for the one in progress.

> In one sentence for the panel: **"FirstLook tells a Meralco crew which piece of equipment to check first after a typhoon, using Meralco's own priority rules, and explains why."**

2 The ranking rules

Every interruption is ranked by the same hierarchy. It comes from the team's policy records and the test file, and it is checked in this order. The first two steps decide whether a job can be worked on at all; the rest decide the order.

1	**Safety** Can the crew work there?	A job marked **unsafe** for the crew is put on hold, whatever else is true about it. The mark comes from the case data. FirstLook never sends a crew to a site marked unsafe.
2	**Flood clearance** Is restoration allowed?	Equipment that is flooded and not yet cleared for restoration is put on hold. An **inspection** may still go ahead, because inspecting is not re-energising. Some hospitals are cleared for **part** of the building (for example floors 3 to 6).
3	**Customer tier** Who is waiting?	Critical customers come before community services, which come before residential customers. The tier always wins over the numbers below it.
4	**Customers restored** How many get power back?	Within the same tier, the job that restores more customer accounts goes first.
5	**Customer reports** Only to break a tie	When the customers restored are equal, the job with more linked outage reports goes first. Reports never move a job past steps 1 to 4.
6	**Shortest driving ETA** The last tie-break	When everything above is equal, the job the crew can reach soonest goes first.

Critical
Hospitals, evacuation centers, a school serving as an evacuation center.

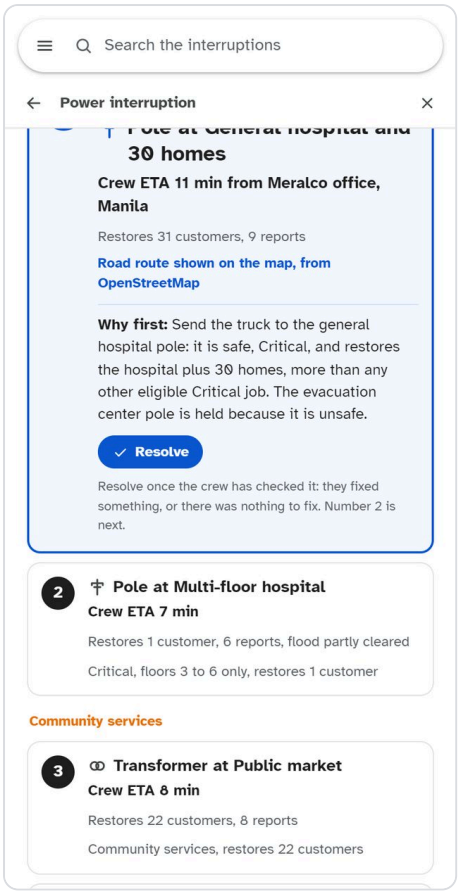
Community services
Public markets, schools, barangay halls.

Residential
Homes, streets, apartment blocks.

Hold the crew. If no job passes steps 1 and 2, the answer is *Hold the crew*: wait, and re-check when a road is made safe or a flood is cleared.

One truck, one job. One crew with one truck does one job at a time. When it arrives it inspects; if there is a problem it fixes it, and if there is nothing to fix it moves on. Both end with *Resolve*.

Worked example: two hospitals in Sampaloc, Manila



Case 14 has ten pieces of equipment. Three serve Critical customers:

- **General hospital and 30 homes.** Safe. Restores 31 customers.
- **Multi-floor hospital.** Flooded on its lower floors; floors 3 to 6 are cleared. Safe. Restores 1 customer.
- **Evacuation center.** Not safe for the crew.

Step 1 puts the evacuation center on hold. Both hospitals pass steps 1 and 2 and are both Critical (step 3), so step 4 decides: **31 customers beats 1**. The general hospital is number 1 and the multi-floor hospital is number 2. That holds even though the multi-floor hospital is closer and has fewer reports.

After the Critical jobs come the community services: the public market (22 customers), the school (12) and the barangay hall (7). Then the residential areas. A flooded line to a residential area waits on hold.

How customer reports are weighed

Reports are evidence, not orders. Two draft policy records written for this build say so, and Liwa reads them every time:

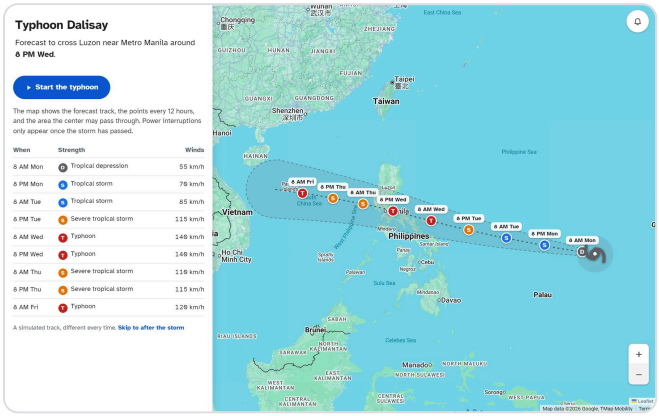
- Reports never override safety, flood clearance, the tier or the customers restored. Their number only breaks a tie after customers restored (step 5).
- **One report does not move a job.** A few busy residential jobs carry exactly one alarming report that nobody else confirms (someone thinks they saw a spark). Liwa notes it for the crew to confirm on arrival, and the order stays as it is.
- What reports *say* is read for the crew: hazards to confirm on arrival, and people who depend on power (oxygen, dialysis, nebulizers). Three or more separate people describing the same hazard at a site marked safe is a reason to re-check its safety, not to re-rank it.

These two records are this project's own draft, not Meralco's or the team's supplied policy. Whoever owns the hierarchy should review them.

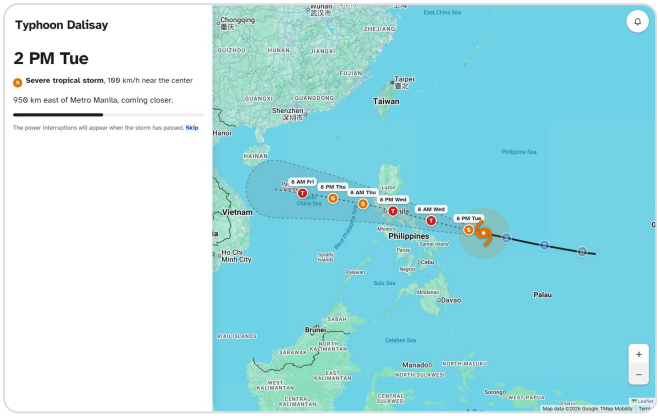
3 The screens, one by one

The app has two parts: the map with its side panel, and the Typhoon records page. This chapter follows the order a person meets them in.

3.1 The typhoon



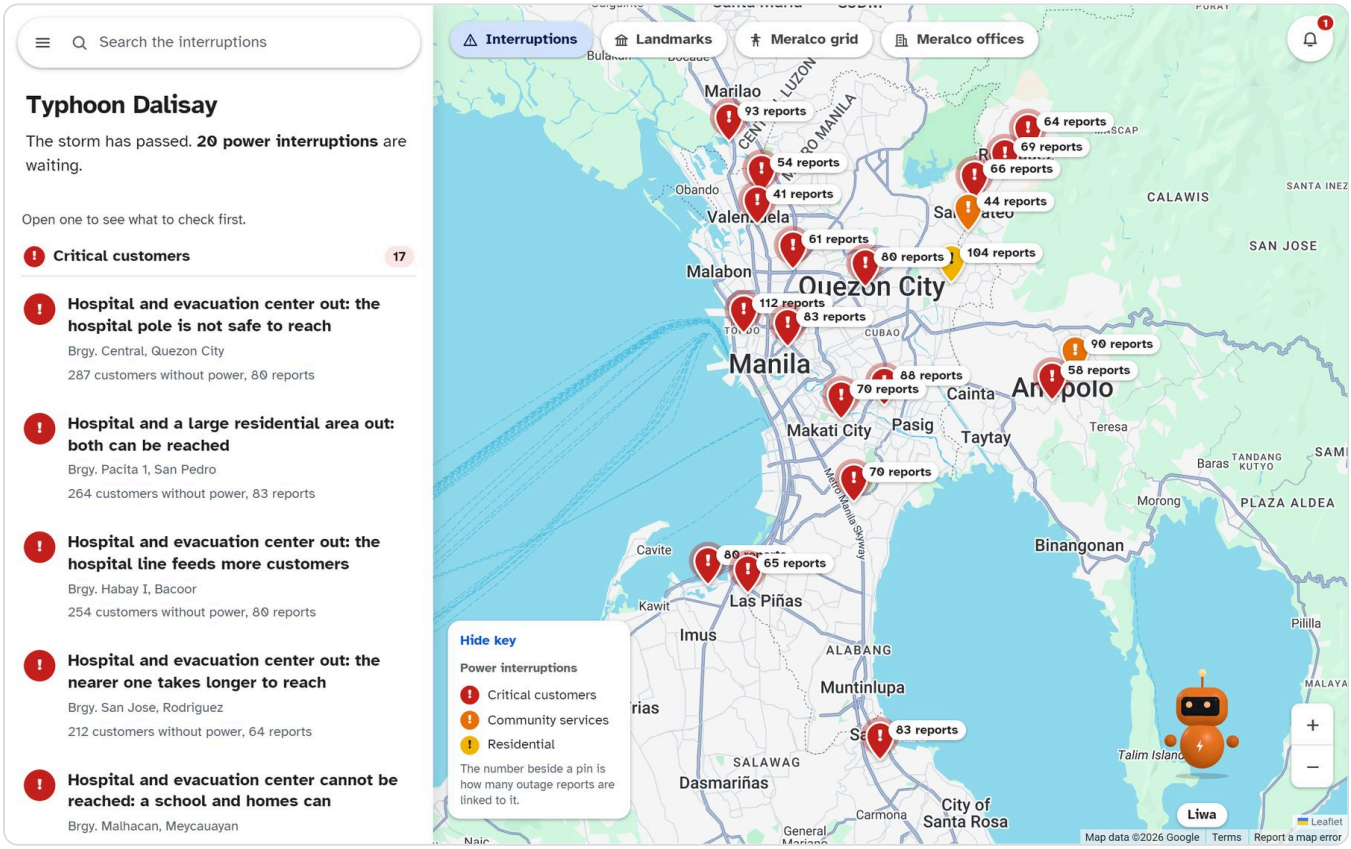
The forecast: a point every 12 hours with its strength, and the area the center may pass through.



After *Start the typhoon*, the storm crosses Luzon near Metro Manila in about 17 seconds.

The app opens on a simulated typhoon. Its track is generated in code and is different every time. No interruption is shown until the storm has passed. *Skip* jumps straight to the interruptions, and *Replay the typhoon* in the menu shows it again.

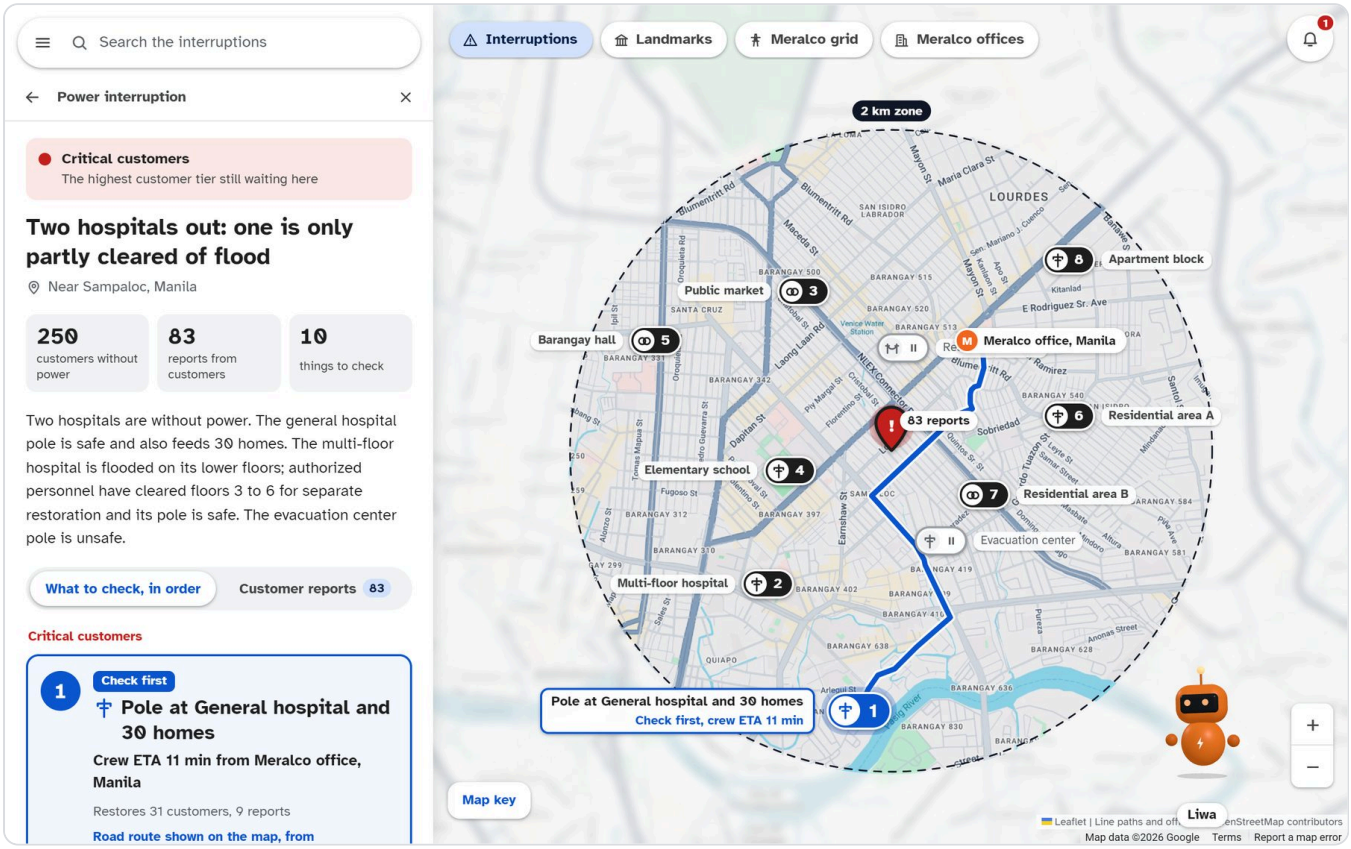
3.2 The interruptions



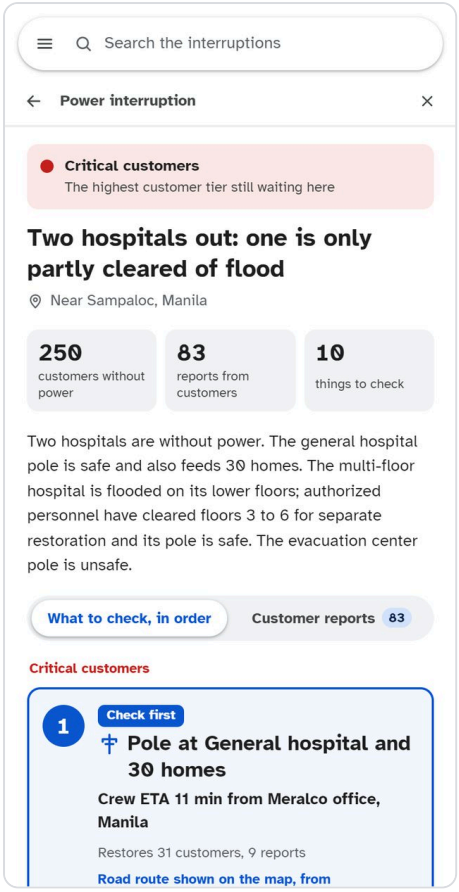
Every interruption as a "!" pin. The side panel groups them by the highest customer tier still waiting.

- The **number beside a pin** is how many customer **outage reports** are linked to that interruption. It is reports, not people or households.
- The pin's color is the highest tier still waiting there: red for Critical, orange for Community services, yellow for Residential.
- The chips at the top switch map layers: *Interruptions*, *Landmarks*, *Meralco grid* (the simulated network) and *Meralco offices*.

3.3 One interruption



One interruption open. Outside its 2 km zone the map is blurred and other pins are hidden. Numbered markers show the order to check; the blue line is the crew's road route to number 1.



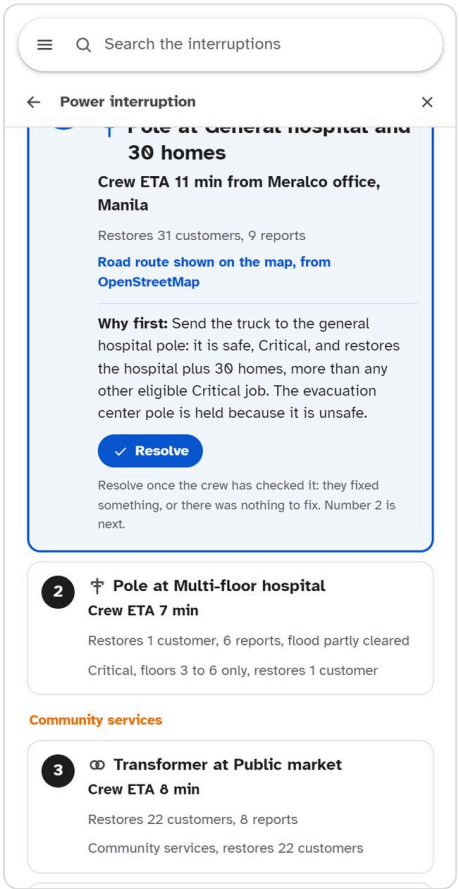
Top of the side panel

- The **title** says what is out and what makes it hard.
- Three numbers: **customers without power**, **reports from customers**, and **things to check**.
- A short description of the situation, from the case.
- Two pills: **What to check, in order** and **Customer reports**.

The ranking, the same on the map and in the panel

- **1, in blue,** is what to check first, with the crew's ETA from the nearest Meralco office and Liwa's reason.
- **2, 3 and onward** come next, under a heading for each customer tier, each with a short reason.
- Each marker carries the equipment's own symbol (pole, transformer or line) beside its number.
- Items **on hold** carry a pause mark and the reason.
- Click any number, on the map or in the panel, to see the route to it instead. Only one route is drawn at a time.

3.4 Number 1 and Resolve



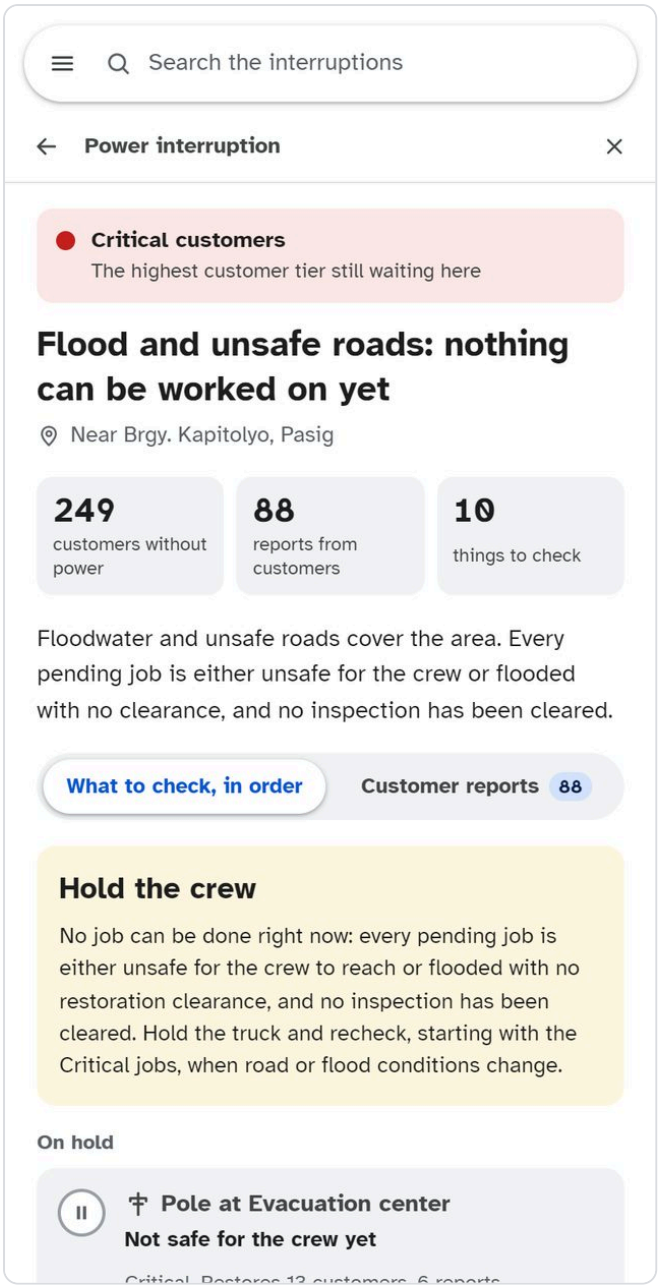
Number 1 has one button: **Resolve**. The crew presses it once they have checked the equipment, whether **they fixed something** or **there was nothing to fix**. Both count as done.

- The job moves to *Done*.
- Number 2 becomes the new number 1, and the route moves with it.
- The count of customers without power drops.
- *Undo* puts it back.

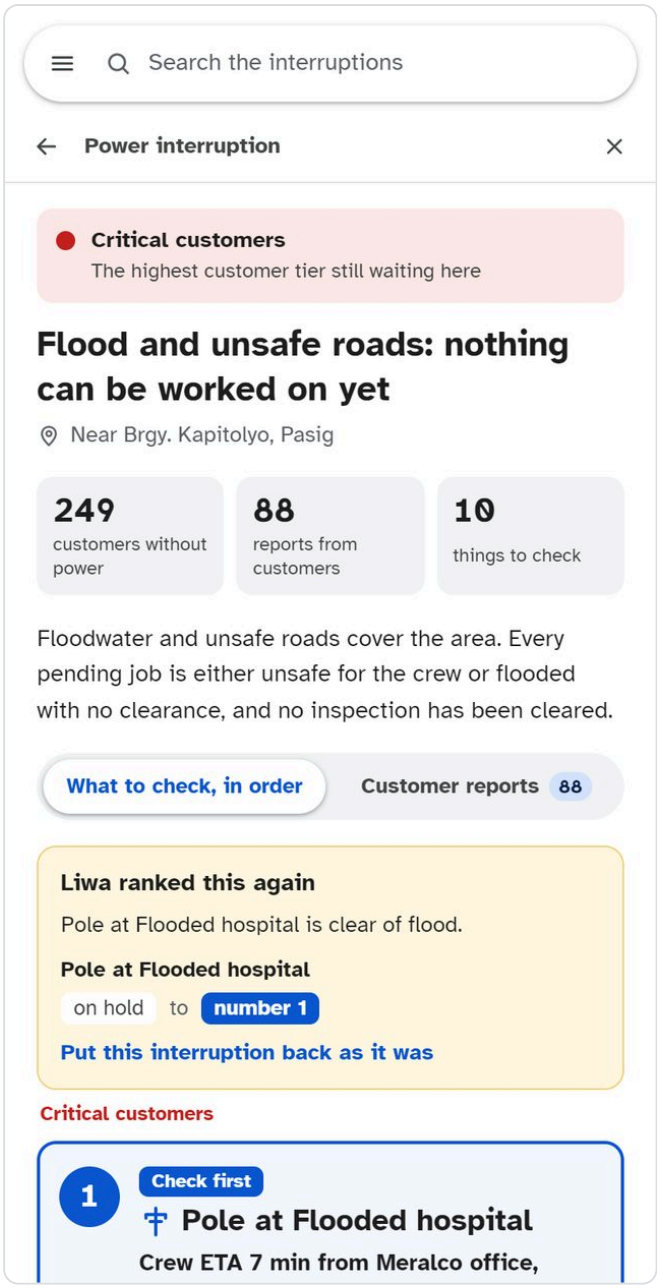
Resolve walks down the order Liwa already gave. It does not ask her again, because taking one job out does not change the order of the rest.

The route says where it came from: Google Maps when the Routes API is enabled on the key, otherwise the public OpenStreetMap router. The ETA shown is always the one in the case, not the route's.

3.5 On hold, and when conditions change



Case 17: flood and unsafe roads hold everything. The answer is *Hold the crew*.



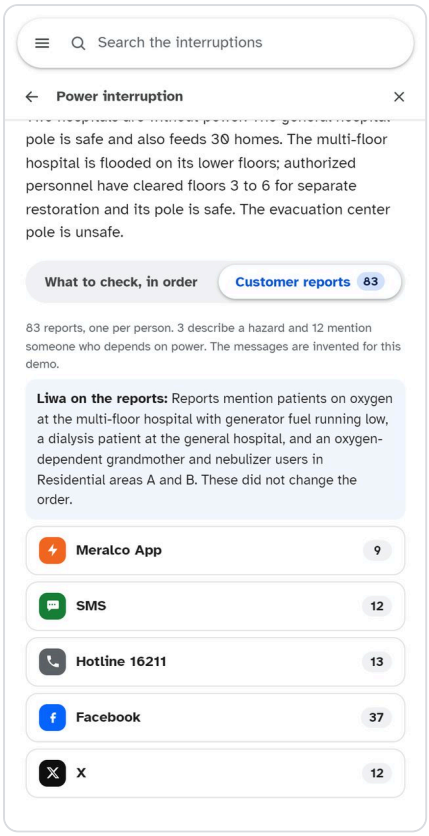
After *Flood cleared* at the hospital, Liwa ranked again: the hospital went from on hold to number 1.

Each job on hold has a button for what can change there: **Now safe for the crew** (a road or site was made safe) or **Flood cleared** (the water is down and restoration is cleared). Pressing one records the change and Liwa ranks the interruption as it now stands, in 6 to 10 seconds. A yellow panel then says what changed and what moved, with her new reason. *Put this interruption back as it was* undoes every change.

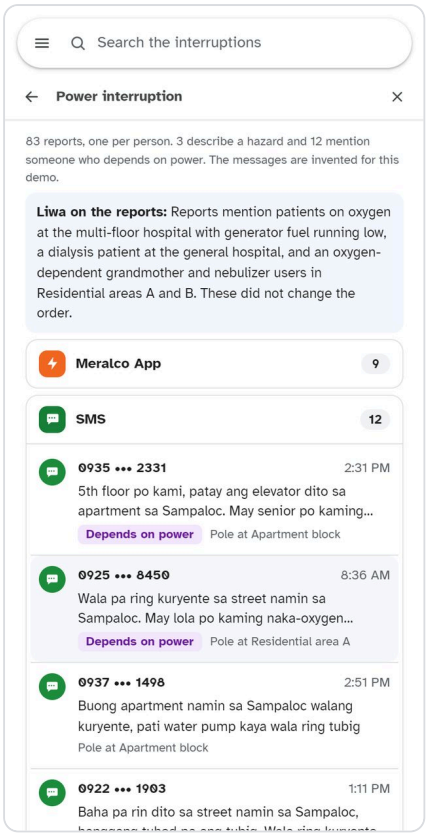
This is where re-ranking matters: a Critical job that was on hold can jump above every community and residential job.

FirstLook does not detect floods or hazards itself. Safety and flood status are fields in the case data. *Flood cleared* and *Now safe for the crew* are a person recording what they were told, for example by a crew or by the authority that clears the area.

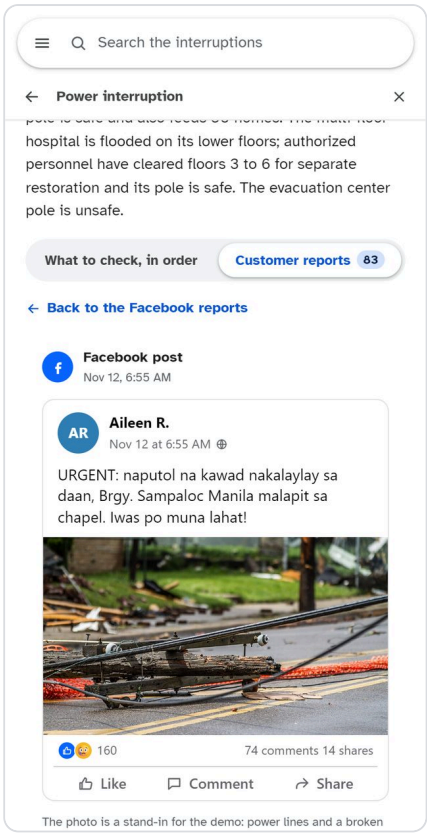
3.6 Customer reports



Reports by channel, with what Liwa took from them on top.



A channel opened: each report, its tags, and the equipment it is linked to.

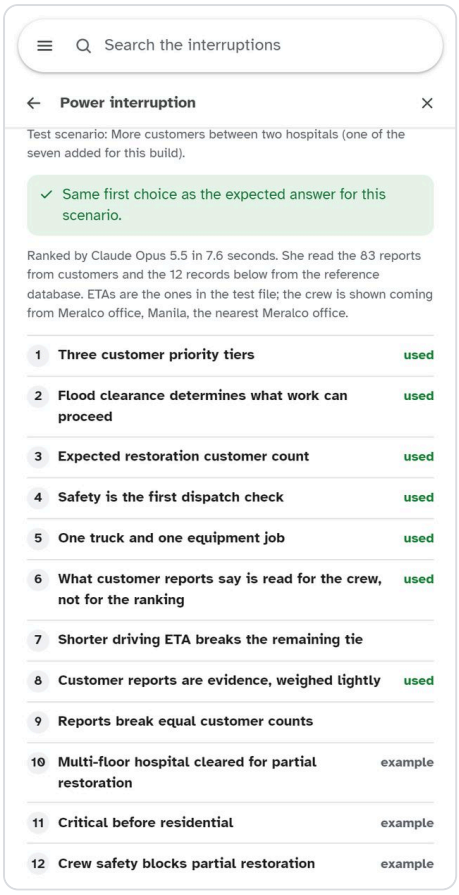


One report shown the way its channel shows it, here a Facebook post.

The second pill lists every report linked to the interruption, by channel: **Meralco App, SMS, Hotline 16211, Facebook and X**. Hazards and people who depend on power are tagged. Opening a report shows it the way that channel looks (a Facebook post or comment, an X post, a text thread, a hotline call ticket, a Meralco App report) and how it counts.

The reports are invented and repeatable: each job gets exactly as many as its report count in the test file, written to fit its facts. The photos are stand-ins from Wikimedia Commons, not pictures of these places.

3.7 How Liwa decided

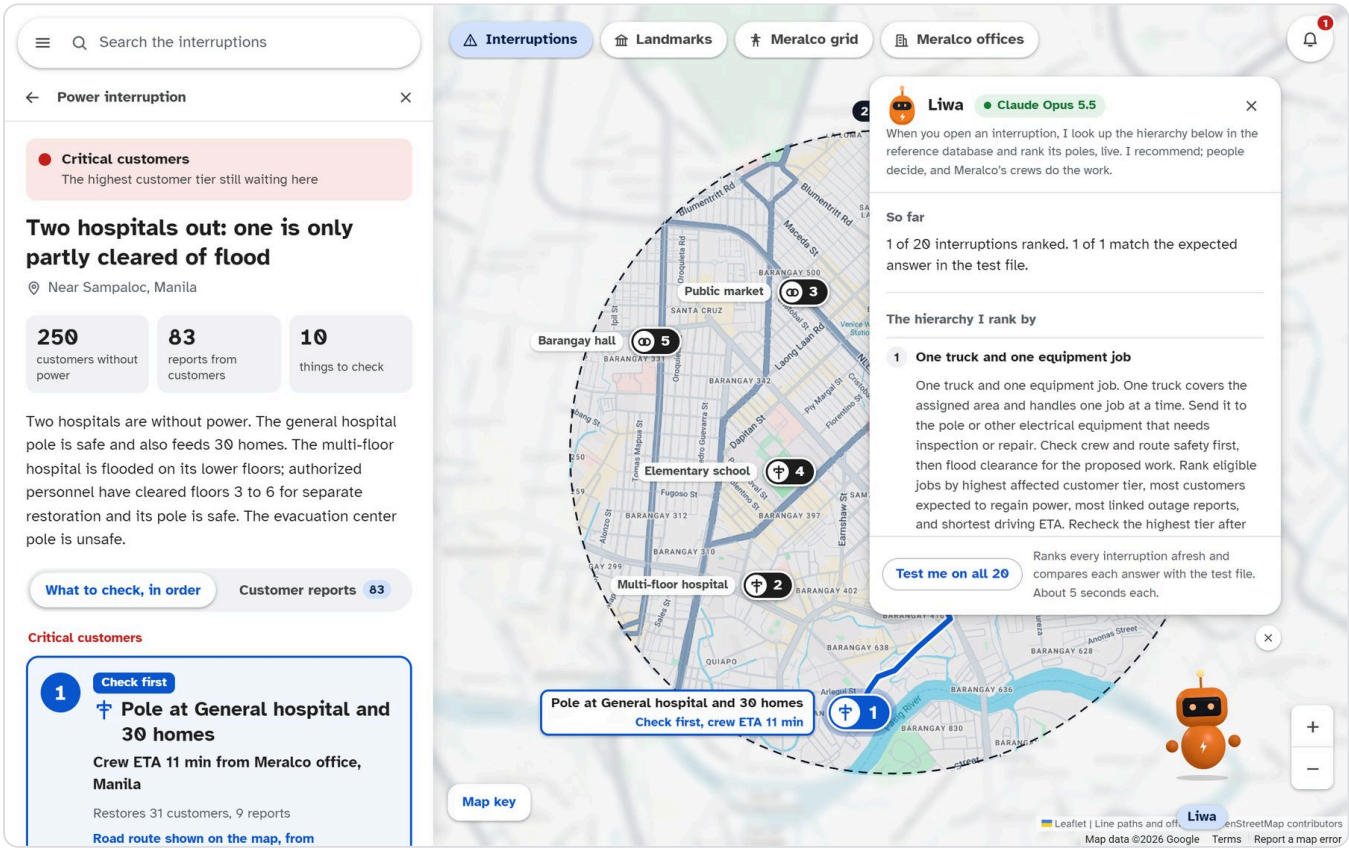


Folded at the bottom of each interruption. It shows:

- The **test scenario** the case comes from.
- Whether Liwa's first choice is the **same as the expected answer** in the test file. After a condition change it is compared with what the written rule gives for the situation as it now stands.
- Which model ranked it, how long it took, and how many reports she read.
- Every **policy record** she was given, marked *used* when she relied on it, and the closest **worked examples**.
- **Ask Liwa again**, to rank it afresh.

This is how a person checks her work: the rules she used and her reason are on screen, next to the answer the test file expects.

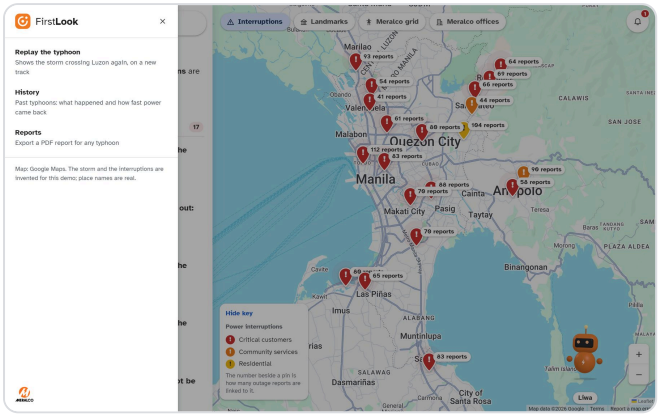
3.8 Liwa



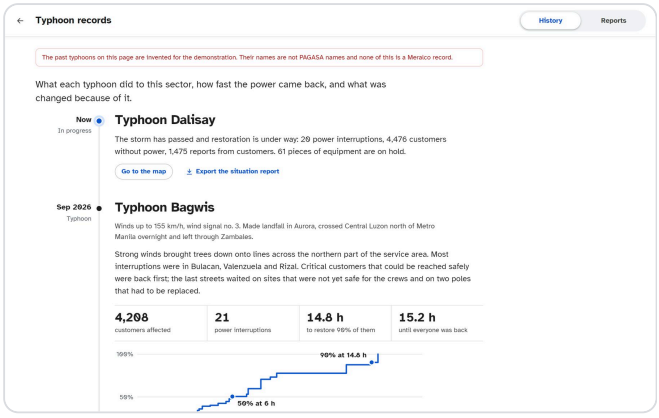
Click the character for Liwa's panel: how many interruptions she has ranked so far, how many matched, and the hierarchy she ranks by.

Test me on all 20 ranks every interruption afresh, one after another, and reports how many first choices match the expected answers. There is nothing to type: Liwa only advises.

3.9 The menu, History and Reports

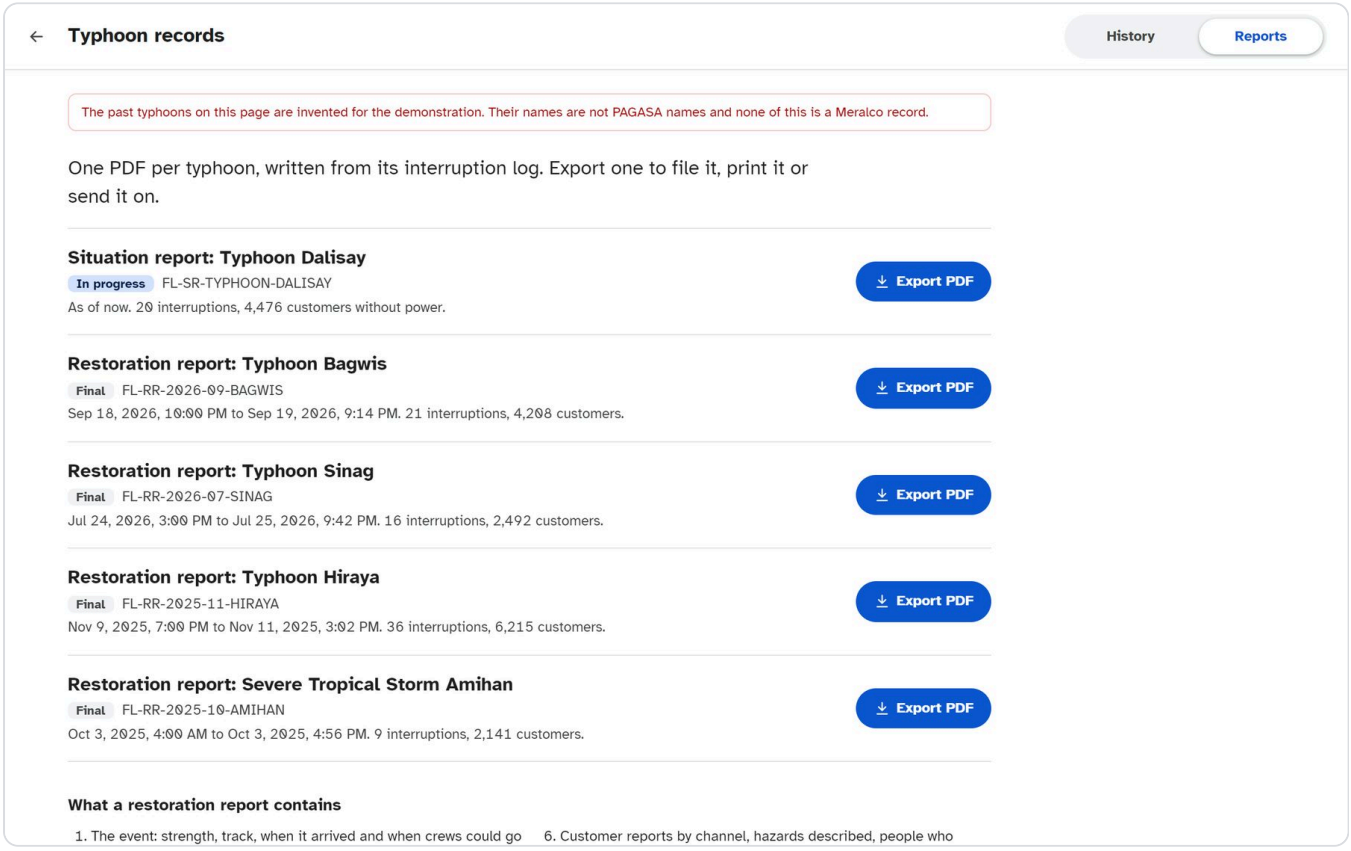


The menu: *Replay the typhoon*, then *History* and *Reports*.



History: past typhoons on a time line, newest first, with the one in progress on top.

History shows, for each typhoon, what happened, four figures (customers affected, interruptions, time to restore 90%, time until everyone was back) and the **restoration curve**: the share of customers with power back, by hours after the all clear. *More about* adds the causes of damage, the impact by customer tier, the customer reports, the holds, and what was changed afterwards.



Reports: one PDF per typhoon. A situation report for the typhoon in progress, a restoration report for each past one.

A **restoration report** has what such a report needs in practice: document number, status, period and area; the event; impact at a glance including customer-minutes interrupted; the restoration curve and milestones; impact by customer tier; causes of damage and work done; customer reports by channel;

safety and holds; how the work was prioritised; crews; the full interruption log; open items; definitions and limits of the data; and a sign-off block (prepared, reviewed, approved). The **situation report** is a snapshot of the typhoon in progress: each interruption and what to check first.

The four past typhoons (Bagwis, Sinag, Hiraya, Amihan) are invented, and their names are not PAGASA names. Only each headline is written by hand; the interruption log is generated, and every figure is counted from that log, so the numbers add up. Every page and every PDF says the data is simulated.

4 How Liwa decides

Liwa is a retrieval-augmented planner. She looks up the rules that apply, hands them with the case to Claude, and her answer is checked before it is shown.

1. The case
Its equipment: tier, safety, flood status, customers restored, reports, ETA. Plus the customer reports linked to each job.

2. Look up the rules
Vector search and keyword search over the reference records, fused into one ranking.

3. Build the brief
Every policy record, most relevant first, and the 3 closest worked examples. Never the case's own example.

4. Claude ranks
Claude Opus 5.5 through the Claude Code CLI returns the action, the order with a reason for each place, a note on the reports, and the records used.

5. Check and show
Only equipment that exists in the case is kept. An unreadable reply is asked for once more; otherwise the written rule answers.

The reference records

- **7 policy records and 13 worked examples** supplied by the team (`backend/reference/rag_documents.jsonl`), for example "Safety is the first dispatch check", "Three customer priority tiers", "Flood clearance determines what work can proceed", "One truck and one equipment job".
- **2 draft policy records** on customer reports, written for this build (`customer_reports.jsonl`).
- They live in one local SQLite file with a full-text index (FTS5) and a vector index (sqlite-vec). The text is embedded with a small model that runs on this computer (`BAAI/bge-small-en-v1.5` , 67 MB).

The written rule, beside her

The test file also has the hierarchy written as plain SQL. FirstLook uses it twice: as a **cross-check** shown beside Liwa's answer, and as the **fallback** when Claude cannot be reached. If Claude Code is missing, signed out or slow, the screen says the answer came from the written rule.

When she ranks

- **When an interruption is opened** for the first time, in about 6 to 10 seconds.
- **When its conditions change** (*Now safe for the crew*, *Flood cleared*).
- **When asked**: *Ask Liwa again*, or *Test me on all 20*.

An answer is kept for as long as the server runs, so the same situation is never ranked twice. Liwa does not watch live feeds by herself. She ranks the data she is given, at the moment it is given.

How well it works

In the last full run, with all the extra equipment and customer reports in front of her, **all 20 first choices matched** the expected answers, and her **whole order was the same as the written rule in all 20**. Two cautions belong with that number:

- The 13 supplied cases had their own worked example hidden from her. The 7 added cases were written for this build with expected answers taken from the written rule, so they show **consistency**, not independent proof.

- She can still misapply a rule. That is why her reason, the records she used and the expected answer are on screen for a person to check.

What it costs

Part	Paid by
Liwa (Claude Opus 5.5)	The Claude account Claude Code is signed in to on this computer. It uses that plan's usage limits. **No API key and no API bill**: the code removes every `ANTHROPIC_*` variable before it calls Claude. Each interruption opened is one call; *Test me on all 20* is 20 calls.
The older message reader (`backend/app/ai.py`)	Would use the paid API, but only if a key were put in `backend/.env`. That line is empty, so it is off. Keep it empty.
Google basemap and directions	The Google Cloud project behind the Maps key in `frontend/.env.local`.
Road router, offices, photos, search model	Free: OpenStreetMap and OSRM, Wikimedia Commons, and a model that runs locally.

5 The 20 test cases

Cases 1 to 13 were supplied by the team in `meralco_demo.sqlite`. Cases 14 to 20 were added for this build, in places that had no interruption yet, to cover situations the first thirteen do not. Every case has ten or eleven pieces of equipment to rank.

Case	Title on screen	What it tests	Where	Expected first answer
1	Hospital and evacuation center out: the hospital pole is not safe to reach	Safety first	Quezon City	**Evacuation center**
2	Hospital and a large residential area out: both can be reached	Critical before residential	San Pedro	**Hospital**
3	Hospital and evacuation center out: the hospital line feeds more customers	More customers restored	Bacoor	**Hospital and 40 homes**
4	School and barangay hall out, with the same number of customers	Reports break the tie	Antipolo	**Barangay hall**
5	Hospital and evacuation center out: the nearer one takes longer to reach	Farther pole is faster	Rodriguez	**Evacuation center**
6	Hospital and evacuation center cannot be reached: a school and homes can	Move down to the community tier	Meycauayan	**School**
7	Critical sites already restored: residential streets still out	Move down to the residential tier	San Mateo	**Residential area A**
8	Every site is unsafe to reach	No safe jobs	Valenzuela	**Hold the crew**
9	Two schools out: one is sheltering evacuees	School serving as an evacuation center	Bacoor	**School A, serving as an evacuation center**
10	Flooded hospital: its upper floors are cleared for power	Partial restoration of a multi-floor hospital	Antipolo	**Multi-floor hospital**
11	Flooded hospital: no clearance to restore it yet	Hospital on flood hold	Rodriguez	**Evacuation center**
12	Hospital cleared for power, but the road to its pole is unsafe	Crew safety blocks partial restoration	San Mateo	**Evacuation center**
13	Flooded hospital: only a pole inspection can be done	Safe inspection while restoration is held	Valenzuela	**Flooded multi-floor hospital (inspection)**
14	Two hospitals out: one is only partly cleared of flood	More customers between two hospitals	Sampaloc, Manila	**General hospital and 30 homes**
15	Hospital and evacuation center flooded: a market and a school can be worked on	Flood holds every Critical job	Tondo, Manila	**Public market**
16	Evacuation center out, and a flooded hospital that needs an inspection	Restoration before inspection	Makati	**Evacuation center**

17	Flood and unsafe roads: nothing can be worked on yet	Flood and safety hold everything	Pasig	**Hold the crew**
18	Only homes left: an apartment tower and several streets	Residential only: most customers first	Marikina	**Apartment tower**
19	Market and barangay hall out: same customers, same reports	Driving time decides between community services	Taguig	**Public market**
20	One shelter restored: a hospital's upper floors and a second shelter waiting	Reports decide between two Critical jobs	Caloocan	**Multi-floor hospital**

What was added, and why

- **More equipment in every case.** The supplied cases had one to four poles each. Six to nine more jobs were added to each (poles, transformers and line sections serving markets, schools, barangay halls and homes; some unsafe, flooded or already done), so there is a real order to show. They were chosen so that **no expected answer changes**, and a test checks that.
- **Seven more cases** (14 to 20) for situations the first thirteen do not isolate: a choice between two hospitals, every Critical job on flood hold, restoration against inspection, flood and safety holding everything, residential only, a tie decided by driving time, and a tie decided by reports.
- **Plain titles.** The test file names each case by the rule it tests ("Safety first"). The title on screen says what is out and what makes it hard. The test name is still shown under *How Liwa decided*.

All additions live in `backend/cases/extra_poles.json` and are added in memory only; the team's sqlite file is never changed. Delete that file, or set `FIRSTLOOK_EXTRA_POLES=off`, to run the cases exactly as supplied.

6 What is real, what is drawn, what is invented

Worth knowing before the panel asks. Given comes from a data source. Drawn is placed on the map to make it readable. Invented was made up for the demo.

Data	Kind	Where it comes from
Each job's tier, safety, flood status, customers restored, report count, driving ETA and distance	Given	The test file (`meralco_demo.sqlite`), plus the equipment and cases added in `extra_poles.json` . FirstLook does not recompute any of it.
Where an interruption sits on the map	Drawn	The test file has no coordinates. Each case is drawn in a real barangay, and its equipment is spread inside the 2 km zone.
Meralco offices	Given	OpenStreetMap volunteers' map, not Meralco's own list. The crew is drawn coming from the nearest one.
Road route	Given	Google Maps directions when the Routes API is enabled on the key; otherwise the public OpenStreetMap router (OSRM); otherwise a straight dotted line. Used for orientation only; the ETA is the case's.
The network (Meralco grid chip)	Invented	One substation and 44 pieces of equipment. The lines follow real roads but are not Meralco's lines.
Customer reports and their photos	Invented	Written to fit each job's facts, the same every time. Photos are stand-ins from Wikimedia Commons.
The typhoon track	Invented	Generated in code, different every run.
Past typhoons and their reports	Invented	Four events with generated interruption logs; every figure is counted from the log.
Policy records and worked examples	Given	The team's reference file, plus two draft records on customer reports written for this build.
Basemap and place names	Given	Google Maps with the team's key; OpenStreetMap when there is no key.

"Customers without power" is the sum of the customers each unfinished job would restore, as given in the cases. It counts customer accounts, not people.

7 How it is built and how to run it

The two halves

Part	Built with	Does
Frontend	React 18, Vite, Leaflet, Google Maps basemap, plain CSS	The map, the side panel, the typhoon, Liwa's character, the Typhoon records page. Resolve is kept in the browser.
Backend	Python, FastAPI, SQLite (FTS5 and sqlite-vec), fastembed, fpdf2	Reads the cases, writes the customer reports, runs Liwa, records condition changes, builds history and the PDF reports.
Planner	Claude Opus 5.5 through the Claude Code CLI	Writes Liwa's ranking and reasons. The written SQL rule is the fallback.

The backend's endpoints

Endpoint	What it returns or does
`GET /api/cases`	Every interruption with its equipment, place, expected answer, the written rule's answer and Liwa's latest decision
`GET /api/cases/{id}/decision`	Liwa's ranking for one interruption (`?force=true` asks again)
`GET /api/cases/{id}/reports`	The customer reports linked to it
`POST /api/cases/{id}/conditions`	Records *Now safe for the crew* or *Flood cleared*
`POST /api/cases/{id}/restart`	Puts one interruption back as it was
`GET /api/history` , `/api/history/{id}`	Past typhoons, and the one in progress
`GET /api/history/{id}/report.pdf`	The PDF report (`current` gives the situation report)

The files that matter

Path	Contents
`backend/cases/meralco_demo.sqlite`	The 13 supplied cases, expected answers, and the hierarchy as SQL views
`backend/cases/extra_poles.json`	Extra equipment, the seven added cases, and plain titles
`backend/reference/*.jsonl`	Liwa's policy records and worked examples
`backend/app/liwa.py`	Liwa: search, the call to Claude, the checks, the fallback
`backend/app/cases.py` , `case_reports.py`	The cases as they stand, and their customer reports
`backend/app/history.py` , `report_pdf.py`	Past typhoons and the PDF reports
`frontend/src/App.jsx`	The app: typhoon, then the interruptions
`frontend/src/panel/` , `map/`	The side panel; the map, markers, route and zone
`frontend/src/records/` , `assistant/`	History and Reports; Liwa's character and panel

Run it

Backend (Python 3.11 or newer), in one terminal:

```
cd backend
python -m venv .venv
.venv/Scripts/python -m pip install -r requirements.txt
.venv/Scripts/python -m uvicorn app.main:app --port 8000
```

Frontend (Node 18 or newer), in a second terminal:

```
cd frontend
npm install
npm run dev
```

Open `http://localhost:5173`. For Liwa's written answers, Claude Code must be installed and signed in (`claude auth status`). For Google Maps, put the key in `frontend/.env.local` as `VITE_GOOGLE_MAPS_API_KEY=...` ; without it the app uses an OpenStreetMap map.

> Condition changes and Liwa's answers are kept in the server's memory. Restarting the backend clears them, and each interruption is ranked afresh when next opened. On this laptop memory is tight; close heavy apps before presenting so the backend is not stopped.

Tests

`cd backend` then `.venv/Scripts/python -m pytest -q` . There are 90 tests, and all pass. They never call Claude or the embedding model. Among other things they check that the written rule still gives every expected answer, that a bad added row is left out, that the team's case file is never written to, that a case never sees its own worked example, that each job gets exactly as many customer reports as its count, and that every figure in the history adds up.

8 Questions the panel may ask

Short answers that stay true to what the system does.

Where does the location of each pin come from?
In this demo, the test cases have no coordinates, so each interruption is drawn in a real barangay and its equipment is spread inside a 2 km zone. The map is a drawing aid; the facts that drive the ranking (tier, customers, ETA) come from the case data.

In real use, the location would come from Meralco's own records: each pole, transformer and line section has an equipment ID with a location, and each customer report is tied to an account's service address and the equipment that feeds it.

Does Liwa work in real time?
She ranks live, at the moment an interruption is opened (6 to 10 seconds), and again whenever its conditions change. She does not watch live feeds by herself; she ranks the data she is given when it is given. Connected to live outage and crew data, the same step would run on every update.

The rule is already written in SQL. Why use AI?
The SQL rule gives an order from the numbers. Liwa adds three things: a reason in plain words for every place in the order; reading what customers actually wrote, so the crew learns about hazards and people who depend on power; and the policy records she used, so a person can check the decision. The SQL rule stays as the cross-check and the fallback.

What if Claude is down or the internet is out?
The answer comes from the written rule in the case file, and the screen says so. The ranking never stops.

How accurate is it?
All 20 first choices matched the expected answers in the last full run, and her full order matched the written rule in all 20. The 13 supplied cases had their own worked example hidden. The 7 added cases show consistency with the rule, not independent proof.

Can it detect floods or unsafe sites?
No. Safety and flood status come from the case data. *Flood cleared* and *Now safe for the crew* are a person recording what they were told. Customer reports that describe a hazard are flagged for the crew to confirm, not used to change the ranking.

Will one angry post push a job up the list?
No. Reports only break a tie after customers restored. One unconfirmed report is noted for the crew and the order stays the same; the demo includes such reports on purpose.

What does Resolve mean?
The crew has checked number 1: they fixed something, or there was nothing to fix. Either way it is done, and number 2 is next.

Where does the ETA come from?
From the case data. The drawn road route is for orientation and does not change the ETA.

Does it cost money to run?

There is no API bill. Liwa uses the Claude account signed in to Claude Code on this computer, within that plan's usage limits. The Google map uses the team's Maps key; everything else is free.

Is this Meralco's data?

No. Everything is simulated: the cases, the network, the reports and the past typhoons. Real places are used only so the map is familiar.

Who makes the final decision?

A person. FirstLook recommends and explains; dispatchers decide and crews do the work. Nothing is sent to a crew automatically.

9 Limits

What FirstLook does not do yet, stated plainly.

- Everything is simulated. The 20 cases are test scenarios, not field records.
- The 20 of 20 result is on these 20 cases only. It says nothing yet about situations the policy records do not cover.
- Liwa's answer is written by Claude from the retrieved records and can still misapply them. Her reason, her records and the expected answer are shown so a person can check.
- FirstLook does not detect floods or hazards, and does not compute travel times. Safety, flood status and ETAs come from the case data.
- Driving times are from the office in the case data, not from where the crew happens to be standing after a job.
- Resolve is kept on screen only. It is not saved, and the situation report does not include it.
- Condition changes and Liwa's answers are lost when the server restarts. When a site with no way in is made safe, a fixed stand-in driving time (5 to 17 minutes) is used.
- The customer reports are invented to fit each job, so they agree with the case by construction. Real reports are messier (wrong places, duplicates, rumours), and that has not been tested.
- The past typhoons are generated. They show what a record and a report would look like, not how FirstLook performed in a real storm. Reports cover one sector and give no system-wide reliability figures (SAIFI, SAIDI).
- The network on the map is invented, and the Meralco offices come from a volunteer map.
- Google directions need the Routes API enabled on the key's Google Cloud project; until then the OpenStreetMap router draws the route.
- A written answer needs internet and a signed-in Claude Code, and counts against that account's usage limits.